Technical Design Document

Cubli

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1 Introduction

1.1 Background & Problem Statement

The Cubli is a reaction-wheel inverted pendulum: a cube-shaped platform that balances on an edge or a corner using only internal actuation, with no external supports or contact forces beyond the single line or point of contact with the ground. The concept was introduced by the Institute for Dynamic Systems and Control (IDSC) at ETH Zurich [1, 2], whose original prototype (15 × 15 × 15 cm) mounts one brushless DC motor and flywheel on each of three orthogonal faces. Commanding a change in wheel speed produces, by conservation of angular momentum, an equal and opposite reaction torque on the cube body; braking a wheel abruptly transfers its stored angular momentum to the body in a near-impulsive event, which is what allows the cube to *jump up* from resting on a face into a balance on an edge or corner.

Dynamically, the system is an *underactuated*, nonlinear, unstable-equilibrium control problem: three actuators (the wheels) must stabilize a body with more degrees of freedom than actuated inputs, about equilibria (edge and corner balance) that are open-loop unstable and increasingly coupled across axes as the contact geometry shrinks from a line (edge) to a point (corner). This is structurally the same problem as the classical inverted pendulum, extended from a single rotational axis to a fully three-dimensional, momentum-exchange-actuated body — which is why the Cubli has become a standard benchmark platform in the controls community over the past decade.

The problem this project addresses is therefore twofold: (i) derive and validate a dynamic model and controller capable of stabilizing this underactuated system about its unstable equilibria and driving it through commanded reorientations, and (ii) realize this in hardware as a modular, largely parametric design, since final actuator and sensor specifications are not yet confirmed at the time of writing. The target capability set, in order of increasing difficulty, is edge balance, corner balance, controlled rotation, and walking (Section 1.3).

1.2 Purpose, Application & Mission Context

The Cubli is not presented here as a balancing novelty but as a terrestrial, gravity-loaded demonstrator of **momentum-exchange attitude control** — the same physical mechanism that governs how real spacecraft point themselves. Nearly every three-axis-stabilized satellite holds and changes its attitude using reaction wheels: flywheels whose commanded speed change produces a reaction torque on the spacecraft body, exactly as in the Cubli. Spacecraft typically carry three or four such wheels (often in a pyramidal arrangement for redundancy) — directly analogous to the Cubli's three orthogonal wheels — and this reaction-wheel approach is the standard for CubeSats and small satellites in particular, making the Cubli representative of a real, currently-flown hardware class rather than an idealized toy problem. Reaction wheels are distinct from control moment gyros (CMGs), which gimbal a constantly-spinning wheel instead of changing its speed to produce much larger torques [3]; positioning the Cubli explicitly as a *reaction-wheel*, not CMG, demonstrator keeps the application claim precise.

Framed as an ADCS problem, the Cubli's core behaviors map directly onto a spacecraft's day-to-day pointing task:

- **Edge / corner balance** → regulation to a commanded reference attitude and disturbance rejection — holding a pointing setpoint against perturbations.
- **Controlled rotation** → a slew maneuver: reference tracking to a new commanded attitude, exactly as a satellite re-aims a payload or antenna.
- **Jump / walk** → momentum-exchange hopping mobility, flight-proven on small bodies

where wheeled traction is impossible. JAXA's Hayabusa2 mission deployed the MINERVA-II1 rovers [4] and the MASCOT lander [5] on asteroid Ryugu (2018), both of which move by spinning up and braking an internal flywheel or eccentric mass to hop across the surface, with MASCOT additionally self-righting the same way. Walking, as a sequence of controlled, directional hops, is a genuine step beyond a single ballistic hop and is an active research frontier (e.g. ETH's SpaceHopper [6]).

This gives the project two complementary application narratives rather than one: it is simultaneously a low-cost, hardware-in-the-loop testbed for standard spacecraft *attitude determination and control* (ADCS), and a ground analog for the momentum-exchange *surface mobility* now used on small-body missions. Framed against a companion star-tracker project, which answers *"where am I pointing?"* (attitude determination), the Cubli answers *"drive me to that reference and hold it"* (attitude control) — the two halves of a complete ADCS loop.

1.3 Objectives

The project targets four behaviors, deliberately ordered as a difficulty ramp in which each milestone is a control precondition for the next:

1. **Edge balance** — stabilize the cube on a single edge, the least unstable equilibrium (contact is a line, one primary axis of instability).
2. **Corner balance** — stabilize the cube on a single vertex, the fully unstable equilibrium (point contact, all three axes coupled and unstable simultaneously).
3. **Controlled rotation** — execute a commanded slew between equilibria (e.g. edge to edge, or in-place reorientation), demonstrating reference tracking rather than pure regulation.
4. **Walking** — chain repeated, directional controlled falls between edges/corners into a net translation across a surface, re-stabilizing after each transition.

These map one-to-one onto the applications of Section 1.2: (1)–(2) demonstrate pointing regulation and disturbance rejection of increasing difficulty; (3) demonstrates slew/reference-tracking; (4) demonstrates momentum-exchange mobility of the kind flown on Hayabusa2. Milestones (1)–(3) are treated as committed deliverables for this design; walking (4) is scoped as the stretch objective, contingent on hardware confirmation and the jump-up feasibility check (angular momentum required to tip the cube vs. available motor/wheel torque), to be resolved once final components are known.

2 Project Organization

2.1 Team Organization

The project is developed by a multidisciplinary team, with responsibilities distributed across the main technical domains of the Cubli: mathematical modeling, control, mechanical/CAD design, and system integration. Table 1 summarizes each member's primary role, and the detailed responsibilities are outlined below.

Table 1: Team members, contact details and primary roles.

Member	**Contact**	**Primary Role**
Pablo Urioste Alarcón	pb.urioste4@gmail.com +34 654 491 174	System integration and lead: digital actuator and sensor coding, embedded software, and integration of the overall design.
Deyan Nikolaev Vlaev	vlaev.dido@gmail.com +359 877 082 499	Mathematical demonstration and derivation of the system dynamics.
Niccolo	—	Control system design and analysis.
Andrea Liang	andrea.liang.2006@gmail.com +31 06 5796 2976	Support on the control problem and structural/mechanical integration.
Suvanna Viriyanti Wu	Viriyanti.wu@gmail.com +39 329 074 7550	CAD modeling and mechanical design.
Athanasia Nikolova	—	CAD modeling, prototype building/assembly, and product presentation (commercial pitch).

Responsibilities.

- **Pablo Urioste Alarcón** — System integration and overall coordination. Handles the coding of the digital actuators and sensors, the embedded software stack, and the integration of the mechanical, electronic and control subsystems into a working prototype.
- **Deyan Nikolaev Vlaev** — Mathematical foundation. Responsible for the analytical demonstration and derivation of the equations of motion and the dynamic model of the Cubli.
- **Niccolo** — Control. Leads the design, tuning and analysis of the control strategy used to balance and maneuver the Cubli.
- **Andrea Liang** — Control and mechanics support. Assists on the control problem and contributes to the structural and mechanical integration of the system.
- **Suvanna Viriyanti Wu** — CAD and mechanical design. Develops the 3D models and mechanical drawings of the structure and reaction-wheel assembly.
- **Athanasia Nikolova** — CAD, prototype building and commercialization. Contributes to the 3D modeling, leads the physical building and assembly of the prototype, and is responsible for presenting and pitching the product.

2.2 Work Breakdown Structure

This plan covers the effort up to the Technical Design Document deliverable on 3 August 2026. The work is grouped into three streams — investigation, requirements and documentation; control and software; and hardware design and build — led by the owners assigned in Table 1. The investigation is kept to the essentials needed to substantiate the design: a focused literature review, the dynamics derivation, and the wheel sizing that follows from it. Table below lists the work packages, their scope, owner and time window.

ID	Work package	Scope	Owner	Window
Investigation, Requirements & Documentation				
I1	Literature & state of the art	Review the ETH Cubli, the one-wheel Cubli, CubeSat ADCS and hopping-mission precedents, and lock the application framing	Suvanna *(+Dejan)*	23–25 Jul
I2	Dynamics investigation	Derive the 1-DoF and 3D equations of motion, linearise and check controllability, and size the wheel from the jump-up energy budget	Dejan *(+Andrea)*	24 Jul–1 Aug
P1	Requirements & planning	Scope, requirements table, milestone staging, roles and risk register	Dejan	23–26 Jul
D1	TDD writing & submission	Each member drafts their section; integrate, insert the S1 evidence, and submit the deliverable	All *(Dejan integrates)*	25 Jul–3 Aug
Control & Software				
C1	Control design (simulation)	Implement the plant model, design and tune the LQR for edge and corner balance, and prototype the state estimator in simulation	Andrea *(+Nicc)*	24 Jul–1 Aug
Hardware Design & Build				
H1	Preliminary CAD & wheel sizing	Envelope study, reaction-wheel design to the inertia target, motor and encoder mounts, and CAD mass-properties export	Neisa, Suvanna	24 Jul–1 Aug
H2	Procurement & 1-DoF rig	Order the long-lead items, bench bring-up of one axis, and build and balance the single-axis test rig	Pablo *(+Neisa)*	23 Jul–2 Aug

2.3 Schedule and Contingency Plan

Figure 1 places the work packages of Section 2.2 on the 23 July–3 August 2026 calendar, and Table 3 states the gate that closes each phase. The critical path runs through the dynamics investigation and wheel sizing, which feed both the control design and the hardware design; the single-axis (S1) rig is the protected item, since its balance result is the experimental evidence in the document. If any modeling or rig task slips, the TDD is still submitted on schedule with simulation-only evidence and the affected result carried as future work.

Table 3: Milestone gates to the deliverable.

Gate	Pass criterion	Date
G1	Long-lead order placed; requirements and plan locked	24 Jul
G2	Dynamics, control design and wheel sizing complete	1 Aug
G3	1-DoF rig balances ≥60 s; plant parameters measured	2 Aug
G4	TDD delivered (deliverable)	3 Aug

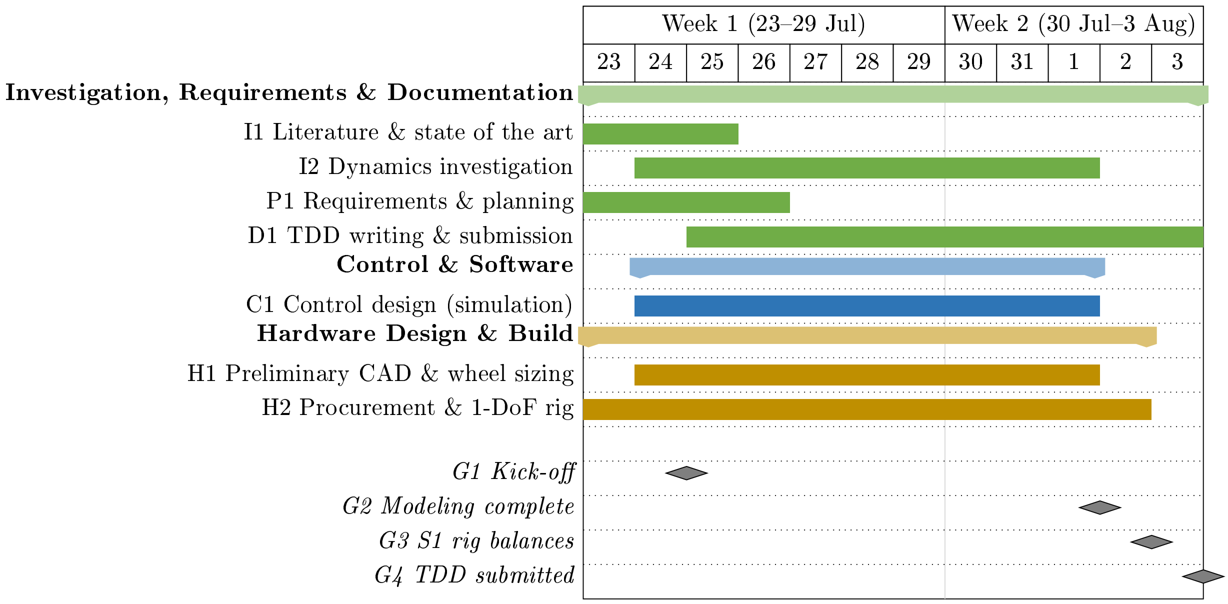

Figure 1: Project schedule to the TDD deliverable, 23 July–3 August 2026. Green: investigation, requirements & documentation; blue: control & software; gold: hardware design & build; grey diamonds: milestone gates.

3 System Analysis & Requirements

3.1 Scope of Work & Use Cases

3.2 Actuation Trade-off — Why Reaction Wheels

3.3 Requirements & Objectives Traceability

Each project objective (Section 1.3) imposes a required capability, which in turn is confirmed by a specific test in the validation plan (Section 6). The mapping below keeps every requirement traceable from motivation to verification.

Table 4: Objective-to-requirement-to-verification traceability.

Objective	Required capability	Verified by
Edge balance	Regulation about a line-contact equilibrium	Sec. 6
Corner balance	Regulation about a point-contact equilibrium, all axes	Sec. 6
Controlled rotation	Reference tracking / slew between equilibria	Sec. 6
Walking (stretch)	Chained jump-up + re-stabilization	Sec. 6

3.4 Coordinate Frames & Conventions

4 Dynamic Modeling & Control

The full Cubli is underactuated and simultaneously unstable about all three axes when balanced on a corner (Section 1.3). The modeling, identification, and control workflow is therefore first validated on a two-dimensional (2D) single-face prototype with one rotational degree of freedom, before being generalized to the 3D cube in Section 4.5.

4.1 2D Reaction-Wheel Pendulum

The 2D prototype is a 3D-printed plate sized to one cube face, with a motor-driven momentum wheel at its center and a braking mechanism at one corner. The plate pivots on a bearing at that corner, constraining motion to a single rotational degree of freedom in a vertical plane.

4.1.1 Equations of Motion

The nonlinear equations of motion are

$$\ddot{\theta}_b = \frac{(m_b l_b + m_w l)\, g \sin\theta_b - T_m - C_b\dot{\theta}_b + C_w\dot{\theta}_w}{I_b + m_w l^2}, \tag{1}$$

$$\ddot{\theta}_w = \frac{(I_b + I_w + m_w l^2)(T_m - C_w\dot{\theta}_w)}{I_w(I_b + m_w l^2)} - \frac{(m_b l_b + m_w l)\, g \sin\theta_b - C_b\dot{\theta}_b}{I_b + m_w l^2}, \tag{2}$$

with motor current-torque relation $T_m = K_m u$. Symbols are defined in Table 5 and used throughout this section.

Table 5: 2D pendulum model variables.

Symbol	Description	Unit
θ_b	Body tilt angle	rad
θ_w	Wheel rotation relative to body	rad
m_b, m_w	Body, wheel mass	kg
I_b, I_w	Body (about pivot), wheel (about motor axis) inertia	$\mathrm{kg\,m^2}$
l	Motor axis to pivot distance	m
l_b	Body CoM to pivot distance	m
g	Gravitational acceleration	$\mathrm{m/s^2}$
T_m	Motor torque	N m
C_b, C_w	Body, wheel damping coefficient	$\mathrm{kg\,m^2/s}$
K_m	Motor torque constant	N m/A
u	Motor current input	A

4.1.2 Equilibria & Linearization

The relevant equilibrium is the upright position $(\theta_b, \dot{\theta}_b, \dot{\theta}_w) = (0, 0, 0)$; its linearization is presented with the control design in Section 4.4.

4.2 Parameter Identification & the 2D Testbed

CAD mass-properties simulation, accounting for print material, infill, and mounted hardware (driver, encoder, MCU, gyroscope/accelerometer, wiring), gives m_b, m_w, I_b, I_w directly, while l and l_b follow from the plate's design geometry. The damping coefficients C_b and C_w cannot be obtained from CAD, since they depend on bearing friction, wiring drag, and motor cogging; nominal values will be estimated for the initial control design, with dedicated testing planned once the prototype is fabricated.

Table 6: Parameter determination method (symbols per Table 5).

Symbol	Determination method
l	Calculated during design
l_b	Calculated during design
m_b	CAD mass properties
m_w	CAD mass properties
I_b	CAD mass properties
I_w	CAD mass properties
C_b	Estimated (future testing planned with prototype)
C_w	Estimated (future testing planned with prototype)
K_m	Motor datasheet

4.3 Jump-Up Feasibility Analysis

Given CAD-derived $I_b, I_w, m_b, l_b, m_w, l$, the wheel speed ω_w required for jump-up is estimated before fabrication, assuming a perfectly inelastic wheel-body collision. Conservation of angular momentum at impact and of energy from impact to edge-standing yield

$$\omega_w^2 = (2-\sqrt{2})\,\frac{I_w + I_b + m_w l^2}{I_w^2}\,(m_b l_b + m_w l)\,g. \tag{3}$$

This value is checked against the motor's achievable speed at the available bus voltage and used to size wheel inertia (radius, rim mass) within the motor's operating range without excessively reducing the balancing recovery angle. Post-fabrication, ω_w is refined experimentally to correct for deviations from the ideal collision assumption.

The braking mechanism uses a servo-driven barrier that strikes a bolt head on the wheel to stop it abruptly. Servo response time bounds the validity of the "instantaneous stop" assumption; a slower stop yields a lower effective post-impact body velocity than predicted, compensated through the same tuning process.

4.3.1 Torque-Limited Wheel-Inertia Target

The relation above assumes an instantaneous, lossless momentum transfer. For preliminary sizing, the achievable motor speed is treated as fixed and the required per-wheel inertia is derived from a momentum budget that additionally accounts for finite brake torque. This gives an ideal per-wheel angular momentum

$$h_{w,\mathrm{ideal}} = \sqrt{\eta}\,\frac{\sqrt{2g\,\lambda\,\kappa}}{n}\,ML^{3/2}, \tag{4}$$

where M, L, g are cube mass, edge length, and gravity, η is an energy margin, and κ, λ, n are dimensionless geometric factors set by the contact case (Table 7).

The corner case is more demanding and governs the design.

Table 7: Contact-case geometric factors.

Case	κ (pivot inertia)	λ (CoM rise)	n (momentum transfer)
Edge	$2/3$	$1/\sqrt{2} - 1/2$	1
Corner (governs design)	$11/12$	$\sqrt{3}/2 - 1/2$	$\sqrt{2}$

A finite brake torque τ_b cannot deliver the impulse instantaneously and must exceed the gravitational tip-over torque $\tau_g = MgL/2$. This loss is captured by a brake-amplification factor

$$\beta = \frac{\tau_b}{\tau_b - \tau_g}, \qquad h_w = \beta\, h_{w,\text{ideal}}, \qquad I_{w,\text{target}} = \frac{h_w}{\omega_{\max}}, \tag{5}$$

where $\omega_{\max}$ is the maximum practical wheel speed. Table 8 gives the resulting target for the baseline cube, which the reaction-wheel design in Section 5.3.3 must meet.

Table 8: Torque-limited wheel-inertia target (corner case, baseline cube).

Symbol	Description	Value
M	Cube mass	1.45 kg
L	Cube edge length	180 mm
η	Energy margin	1.05
τ_b	Brake torque per wheel	5.0 N m
τ_g	Gravitational tip-over floor	1.28 N m
β	Brake-amplification factor	1.34
$\omega_{\max}$	Max wheel speed	8400 rpm (880 rad/s)
h_w	Required angular momentum	0.277 $\text{kg m}^2/\text{s}$
$I_{w,\text{target}}$	**Target wheel inertia**	$\mathbf{3.15 \times 10^{-4}}$ **kg m²**

4.4 Control Design on the 2D Model

4.4.1 LQR Baseline Control

Linearizing about the upright equilibrium $(\theta_b, \dot\theta_b, \dot\theta_w) = (0, 0, 0)$ gives

$$\dot x(t) = Ax(t) + Bu(t), \qquad x := (\theta_b, \dot\theta_b, \dot\theta_w), \tag{6}$$

with A, B built from the parameters in Table 5. Zero-order-hold discretization at a rate set by the open-loop unstable pole, with a safety margin, gives $x[k+1] = A_d x[k] + B_d u[k]$. A discrete LQR gain $K_d = (K_{d1}, K_{d2}, K_{d3})$ minimizes

$$J(u) = \sum_k x^{\mathrm{T}}[k]Qx[k] + u^{\mathrm{T}}[k]Ru[k], \tag{7}$$

giving $u[k] = -K_d(\hat\theta_b[k], \hat{\dot\theta}_b[k], \hat{\dot\theta}_w[k])$.

The tilt estimate $\hat\theta_b$ is obtained from a gyroscope and accelerometer mounted at, or as close as mechanically feasible to, the pivot point rather than at the plate's center. This placement is necessary because an accelerometer at radius r from the pivot measures gravity plus the tangential ($r\ddot\theta_b$) and centripetal ($r\dot\theta_b^2$) acceleration components induced by the plate's rotation, which corrupt the naive estimate $\tan^{-1}(-a_x/a_y)$ during fast motion (e.g., jump-up or aggressive recovery); placing the sensor at $r \approx 0$ makes both terms negligible. Tilt is estimated with a complementary filter combining gyro integration (drift-prone, unaffected by the terms above) with accelerometer inclination (drift-free, but unreliable during shocks):

$$\hat{\theta}_b[k] = (1-\gamma)\big(\hat{\theta}_b[k-1] + \dot{\theta}_{b,\text{gyro}}[k]\,\Delta t\big) + \gamma\,\theta_{b,\text{accel}}[k], \tag{8}$$

where $\theta_{b,\text{accel}}[k] = \tan^{-1}(-a_x[k]/a_y[k])$ is the instantaneous accelerometer-only estimate and $\gamma \in (0,1)$ is a small weight favoring the gyro term between corrections.

A constant tilt-sensor offset d, including residual bias from imperfect pivot placement or sensor misalignment, would otherwise appear as a nonzero steady-state wheel speed rather than a corrected angle. A low-pass filter state

$$x_f[k+1] = (1-\alpha)x_f[k] + \alpha\hat{\theta}_b[k] \tag{9}$$

is subtracted from the tilt feedback,

$$u[k] = -K_d(\hat{\theta}_b[k] - x_f[k],\, \hat{\dot{\theta}}_b[k],\, \hat{\dot{\theta}}_w[k]), \tag{10}$$

so d is absorbed by x_f at steady state, requiring no separate calibration stage. This fixed-gain design is the baseline balancing controller; Table 9 summarizes the control variables.

Table 9: Control-design variables (2D and corner-balancing 3D model).

Symbol	Description
x	State vector
A, B	Continuous-time state/input matrices (linearized model)
A_d, B_d	Discrete-time (ZOH) counterparts of A, B
K_d	LQR feedback gain
Q, R	LQR state/input weighting matrices
$J(u)$	LQR cost function
$\hat{\theta}_b, \hat{\dot{\theta}}_b$	Estimated tilt angle, rate (gyroscope/accelerometer)
$\hat{\dot{\theta}}_w$	Estimated wheel rate (encoder)
a_x, a_y	Raw accelerometer readings (tangential, radial axes)
$\theta_{b,\text{accel}}$	Instantaneous accelerometer-only tilt estimate
$\dot{\theta}_{b,\text{gyro}}$	Raw gyro rate measurement
γ	Complementary-filter weight (accelerometer term)
Δt	Control/estimation loop sample period
x_f, α	Offset-correction filter state, filter coefficient
d	Constant tilt-sensor offset
θ_{th}	Tilt threshold, jump-up $\rightarrow$ balancing switch
$K_d^{(i)}$	Scheduled gain at linearization point i (extension)

4.4.2 Gain Scheduling & Mode Transitions

The controller switches from the open-loop jump-up sequence to the LQR law once $|\hat{\theta}_b| < \theta_{\text{th}}$, with hysteresis around θ_{th} to prevent mode chattering. The baseline uses a single fixed gain K_d at the upright equilibrium. If recovery degrades near the edges of the operating range, where the small-angle linearization weakens, gain scheduling will be added: additional gains $K_d^{(i)}$ computed at several linearization points, with the controller interpolating between them based on $\hat{\theta}_b$, reusing the same discrete structure and offset filter.

4.5 Generalization to the 3D Cube

The dynamics, identification workflow, and control approach validated on the 2D model extend to the full three-dimensional cube, where three orthogonal reaction wheels must stabilize the body about its edge and corner equilibria.

4.5.1 Full-Body Equations of Motion

With $R \in SO(3)$ the cube's orientation and $\omega_b \in \mathbb{R}^3$ its body-frame angular velocity, the attitude kinematics are

$$\dot{R} = R\,[\omega_b]_\times, \tag{11}$$

where $[\cdot]_\times$ is the skew-symmetric cross-product matrix. The assembly's angular momentum about the pivot, expressed in the body frame, generalizes the single-axis momentum term of the 2D model:

$$H = I\omega_b + \sum_{i=1}^{3} I_{w,i}\,\omega_{w,i}\,e_i. \tag{12}$$

Euler's equation gives

$$\dot{H} + \omega_b \times H = \tau_g - C_b\,\omega_b, \tag{13}$$

and each wheel obeys

$$I_{w,i}\big(\dot{\omega}_{b,i} + \dot{\omega}_{w,i}\big) = T_{m,i} - C_{w,i}\,\omega_{w,i}, \qquad i = 1,2,3. \tag{14}$$

Table 10 defines all symbols.

Table 10: Full-body (3D) model variables.

Symbol	Description
R	Cube orientation relative to world frame
ω_b	Body angular velocity, body frame ($\in \mathbb{R}^3$)
$\omega_{b,i} = e_i^{\mathrm{T}}\omega_b$	Body angular velocity about wheel axis i
$\omega_w = (\omega_{w,1}, \omega_{w,2}, \omega_{w,3})$	Wheel angular velocities relative to body
e_i	Spin axis of wheel i (face normal)
I	3×3 inertia tensor, body+wheels, about pivot
H	Total angular momentum about pivot
$\tau_g = m\,r_{cm} \times (R^{\mathrm{T}}\hat{g})$	Gravity torque about pivot
m	Total assembly mass
r_{cm}	Body-frame vector, pivot to assembly CoM
$\hat{g}$	World vertical unit vector
C_b	3×3 body damping matrix
$I_{w,i}, C_{w,i}$	Inertia, damping of wheel i
$T_{m,i} = K_{m,i}u_i$	Motor torque, wheel i
$u_i, K_{m,i}$	Motor current, torque constant, wheel i

Restricting motion to a single vertical plane with two wheel torques set to zero recovers the 2D prototype model exactly, with $I_b, l_b, m_b, m_w, I_w, C_b, C_w$ the corresponding single-axis projections of $I, r_{cm}, m, I_{w,i}, C_{w,i}$.

4.5.2 Edge vs. Corner Equilibria & Linearization

Jump-up proceeds in two stages: Flat-to-Edge, where one wheel brakes, followed by Edge-to-Corner, where a second wheel brakes. **Edge balancing** constrains the body to a single axis, so the dynamics reduce to the 2D form, and the 2D LQR baseline and offset filter apply directly, with m_b, l_b, I_b replaced by their edge equivalents.

Corner balancing is genuinely 3D: all three wheels contribute across three rotational DOF. Linearizing about the corner-up equilibrium $(\theta, \omega_b, \omega_w) = (0, 0, 0)$, with $\theta \in \mathbb{R}^3$ the small orientation error, gives the same state-space form as the 2D case:

$$\dot{x}(t) = Ax(t) + Bu(t), \qquad x := (\theta, \omega_b, \omega_w) \in \mathbb{R}^9, \tag{15}$$

with $A \in \mathbb{R}^{9\times 9}$ and $B \in \mathbb{R}^{9\times 3}$ built from I, r_{cm}, and the per-wheel parameters $I_{w,i}, C_{w,i}, K_{m,i}$ (Table 10). The LQR design generalizes directly: K_d becomes a 3×9 gain matrix, using the same discretization, cost function, and offset filter as Table 9, with Q, R weighting all three tilt axes, body rates, and wheel speeds. The mode-transition logic gains a third mode, Edge-to-Corner jump-up, triggered once edge balancing is stable, followed by a second threshold switch into the corner LQR once estimated orientation nears corner-up.

5 Preliminary Hardware Design

This design is preliminary: final actuator and sensor specifications are not yet confirmed, so the mechanical and electrical architecture is kept modular and largely parametric (Section 1.1).

5.1 System Architecture

5.2 Mass and Inertia Budget

Before committing to component geometry, the preliminary design is bounded by three coupled constraints that must be satisfied simultaneously. Sizing to the worst case of each fixes the envelope within which the mechanical and electrical design then iterate.

Volume constraint. Every subsystem—the three orthogonal reaction-wheel/motor modules, the battery, and the electronics stack—must fit inside the 180 mm cube envelope. The binding dimension is the reaction wheel: its outer diameter must clear the internal frame and corner/edge contact features while leaving room for the motor and the servo brake. This caps the usable wheel diameter and, with it, the radius at which ballast mass can be placed—directly limiting the inertia achievable at a given wheel mass (Section 5.3.3).

Mass constraint. The cube is designed to a total mass target of $M = 1.45$ kg. This is not merely a logistics figure: it feeds back into the dynamics, since the gravitational tip-over torque $\tau_g = MgL/2$ and the required jump-up angular momentum both scale with M (Section 4.3.1). A heavier cube demands a more capable brake and a larger wheel inertia, so the mass target and the inertia target are coupled and must be closed together. The preliminary allocation is given in Table 11, with the reaction-wheel set (the one firmly-sized item) committed and the remaining subsystems held as allowances to be refined as components are selected.

Table 11: Preliminary mass budget (per cube). Reaction wheels sized; others allocated.

Subsystem	Allocated mass (g)	Status
Reaction wheels (3×)	395	Sized (Sec. 5.3.3)
Motors (3×)	TBD	Allowance — pending motor selection
Frame / structure	TBD	Allowance — pending CAD
Battery	TBD	Allowance — pending capacity sizing
Electronics (MCU, drivers, IMU, servo)	TBD	Allowance
Wiring / fasteners / margin	TBD	Allowance
Total (target)	**1450**	**Design target**

Inertia constraint (corner-critical). Each reaction wheel must store enough angular momentum at the motor's maximum speed to execute the jump-up maneuver. The two contact cases are not equally demanding: the corner equilibrium requires a larger rise of the center of mass and a less favorable momentum-transfer geometry than the edge equilibrium, so for a common maximum wheel speed $\omega_{\max}$ it demands the greater wheel inertia. Evaluating the torque-limited budget of Section 4.3.1 for both cases (Table 12), the corner case requires 3.15×10^{-4} kg m^2 against 2.85×10^{-4} kg m^2 for the edge —about 10% more. The corner jump-up is therefore the critical condition, and the reaction wheel is sized to it; a wheel that satisfies the corner requirement satisfies the edge requirement with margin.

Table 12: Required per-wheel inertia by contact case, at $\omega_{\max} = 8400$ rpm.

Contact case	h_w (kg m^2/s)	$I_{w,\text{target}}$ (10^{-4} kg m^2)
Edge	0.251	2.85
Corner (critical)	**0.277**	**3.15**

5.3 Mechanical Design

5.3.1 2D Prototype Build

5.3.2 3D Cube Frame

5.3.3 Reaction-Wheel Sizing

The reaction wheel must supply the per-wheel inertia $I_{w,\text{target}} = 3.15 \times 10^{-4}$ kg m^2 derived from the jump-up budget of Section 4.3.1, while remaining light enough not to dominate the cube mass. Rather than adjusting inertia by scaling a solid disc, the wheel is built as a printed *ring plus spokes* whose inertia is then tuned by *ballast*: standard steel hex nuts dropped into hexagonal pockets distributed around the ring. Each pocket both removes plastic and adds steel, so the net inertia gain per pocket is large and the design target is reached by choosing how many nuts to install—a discrete, easily-adjustable knob that also lets the physical wheel be re-tuned after fabrication to absorb parameter uncertainty.

The structure is printed in PET-CF ($\rho = 1290$ kg/m^3): a 120 mm outer-diameter ring of 15 mm radial width and 10 mm thickness, carried by three radial spokes 10 mm wide of the same thickness. The hub and motor interface are neglected in the inertia budget. The ballast is standard M5 steel hex nuts ($\rho = 7850$ kg/m^3, 11 mm across flats, measured mass 3.12 g), seated in blind hexagonal pockets machined at the mid-width of the ring. Accounting for the plastic each pocket displaces, the *net* contribution per installed nut is +2.39 g of mass and $+1.55 \times 10^{-5}$ kg m^2 of inertia at the ring radius.

Table 13 sweeps the outer diameter, solving for the number of nuts needed to reach $I_{w,\text{target}}$ and checking that the pockets physically fit. Small wheels need so many nuts that no plastic wall survives between adjacent pockets; wheels of 140 mm and above reach the target from the bare ring alone (no tuning margin, and heavier). The selected design is **OD = 120 mm with 21 nuts**: it meets the inertia target with a ∼2.5 mm circumferential wall between pockets, splits the inertia roughly evenly between structure (45%) and ballast (55%) so the wheel can be trimmed in either direction, and keeps the three-wheel set to ∼395 g (27% of the 1.45 kg cube).

Table 13: Reaction-wheel outer-diameter trade study. Selected design in bold.

OD (mm)	N	Mass (g)	I_w (10^{-4} kg m^2)	I_w/target (%)	3 wheels (g)	Pocket fit
80	105	300.6	3.17	100.7	901.7	fails — no wall
90	72	229.6	3.16	100.5	688.7	fails — no wall
100	51	187.3	3.25	103.3	561.9	fails — no wall
110	33	152.2	3.22	102.3	456.7	fails — thin wall
120	**21**	**131.5**	**3.31**	**105.2**	**394.6**	**OK (∼2.5 mm)**
130	9	110.8	3.23	102.7	332.4	OK (loose)
140	0	—	3.23	—	—	bare ring exceeds target
150	0	—	4.07	—	—	bare ring exceeds target

Radial fit caveat. The circumferential fit is satisfied at 120 mm, but the *radial* fit is marginal: an 11 mm across-flats pocket in a 15 mm-wide ring leaves only 1.8 mm of wall on each side (dropping to 0.9 mm if a nut is seated corner-out), below the 2 mm minimum-wall target. The ring width should therefore be increased to $\gtrsim 17$ mm, or the nut orientation constrained flats-radial, in the next design iteration. This does not change the inertia budget, only the ring's structural robustness around the pockets.

5.4 Electrical Design

5.4.1 Components

The electronics and software will be built around a microcontroller communicating over a CAN bus with a motor driver that closes the current loop for the selected brushless motor. Wheel speed will be measured with a magnetic encoder on the motor shaft, and body tilt will be estimated from an onboard IMU. Braking will be actuated by a digital servo driven directly from the microcontroller, and a wireless module will provide a telemetry and debugging link. Power will be supplied by a LiPo battery, stepped down through a DC-DC converter for the logic-level electronics, with the motor driver powered directly from the battery bus.

The full parts list is catalogued in Appendix A: the supplied components grouped by subsystem (Table 14), the actuation figures derived from them (Table 15), and the mechanical and consumable items still to be procured (Table 16). The supplied set is an electronics-only BOM; the reaction-wheel rims, the frame and all printed parts are the team's design responsibility, and their procurement is the leading schedule risk (Table 16).

5.4.2 Electrical Diagram

5.5 System Integration

6 Testing and Validation Plan

This plan covers system-level validation. Parameter-identification tests that feed the model directly (swing test, current-step identification) are described with the model they validate in Section 4.2.

6.1 Simulation-First Validation

6.2 Bring-up Sequence

6.3 Sensor Testing

6.4 Milestone Acceptance Criteria

6.5 Control Performance Evaluation

7 Future Work

7.1 NMPC & Warm-Start Control (Future Implementation)

7.2 Autonomous Stand-up & Disturbance Rejection

7.3 Single-Wheel Variant / Extensions

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A Bill of Materials

The components supplied for this project constitute an *electronics* bill of materials: the actuation chain, wheel and attitude sensing, on-board compute, communications and power distribution are specified, but no structural or mechanical hardware is. The reaction wheels, the frame, the motor mounts, the fasteners and every printed bracket are the team's design responsibility, and are treated as such in Section 5. Table 14 lists the supplied items grouped by subsystem; Table 15 collects the actuation figures that are *computed* from those parts rather than read off a datasheet — the torque constant, the achievable torque and wheel speed, and the inertia and stored energy that size the wheel; and Table 16 enumerates what must still be bought or fabricated.

The architecture the BOM implies is a two-level control stack: each mjbots moteus-n1 closes a high-rate field-oriented current loop on its motor, using the MA600 encoder for rotor angle, while the Teensy 4.1 runs the outer attitude-estimation and state-feedback loop over CAN-FD — the same inner-tachometer / outer-attitude layering used in spacecraft reaction-wheel assemblies. A mechanical wheel brake, following the original ETH Cubli [2], provides the near-impulsive momentum transfer required for the jump-up.

Procurement is the single largest schedule risk. The long-lead items are not electronic but mechanical: laser-cut steel wheel rims and frame stock (Table 16) gate the entire build, and a shortfall in the motor count would force the reduced-actuation one-wheel variant [**hofer2023onewheelcubli**]. These items are ordered first.

Table 14: Supplied bill of materials, grouped by subsystem.

Ref	Item	Qty[a]	Key specifications	Function
Actuation				
1	T-Motor MN4006 KV380	3	18N24P outrunner; 380 rpm/V; 68 g; ⌀44.35 mm × 21 mm; 194 mΩ; peak 16 A, 380 W	Reaction-wheel torque source
2	mjbots moteus-n1	3	10–54 V; 100 A peak / 9 A cont.; 15–30 kHz FOC; 5 Mbit/s CAN-FD; 14.6 g	FOC current/torque driver (inner loop)
11	TowerPro MG92B servo	1–3	metal-gear micro servo; ≈0.29 N m (≈3 kg cm) at 6 V; 50 Hz PWM	Wheel brake actuator (jump-up)
Sensing				
3	mjbots MA600 breakout + ring magnet	3	TMR absolute angle; SPI 6 MHz, 3.3 V; 16-bit (65,536 counts/rev); air-gap ≤ 0.5 mm	Wheel angle: commutation + state feedback
10	SparkFun BMI270 6-DoF IMU	1	3-axis accel + 3-axis gyro (no magnetometer); I^2C 400 kHz (Qwiic); ODR up to ≈ 1 kHz	Body attitude reference (tilt + rates)
Compute and Communications				
4	Teensy 4.1	1	600 MHz Cortex-M7 + FPU; native CAN-FD (CAN3); 5 V supply	Flight computer: estimator + control law
5	CAN-FD adapter (Teensy 4.1)	1	CAN-FD transceiver, logic-level → differential; 5 Mbit/s	Bus transceiver for the MCU
6	mjbots mjcanfd-usb-1x	1	USB ↔ CAN-FD, 5 Mbit/s	Bench interface (calibration, tuning)

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Table 14 (continued)

Ref	Item	Qty[a]	Key specifications	Function
7	JST-PH3 CAN cables	≥4	TODO: conductor gauge / current rating not in source; 3-conductor daisy-chain	CAN bus interconnect
8	Seeed XIAO ESP32-C6 + antenna	1	Wi-Fi; 3.3 V UART to MCU; TX burst ≈ 300 mA	Wi-Fi telemetry / command link
9	60.4 Ω 0.5 W resistor	4	split termination, 2 × 60.4 Ω ≈ 120 Ω bus impedance	CAN bus termination
17	4.7 nF capacitor	2	common-mode filter, bus midpoint to ground	CAN split-termination filter
Power				
12	Turnigy 6S LiPo (XT90)	1	22.2 V nom / 25.2 V charged; 3600 mA h; 60C; ≈ 80 W h; ≈ 600 g	Primary energy store
13	LM2596 buck module	1	step-down to 5.0 V; rated 2–3 A (derates without heatsink)	5 V logic rail
14	XT90 connector pair	1–2	rated ≈ 90 A	Battery power connector
15	100 µF 16 V electrolytic	2–4	16 V part — 5 V rail only	5 V-rail bulk reservoir
19	1N5819 Schottky diode	1–2	1 A, 40 V	5 V-rail back-feed protection
Integration				
16	100 nF 50 V capacitor	6–10	local decoupling at each IC	Per-IC supply decoupling
18	Double-sided perfboard	1	15 cm × 9 cm, double-sided	Power / signal distribution board

[a] Quantity per assembled cube; motors, drivers and encoders are 3 each. A fourth actuation set (motor + moteus-n1 + MA600) is retained on the 1-DoF bench rig (S1) throughout the build, so four of each are procured.

Table 15: Derived actuation parameters (computed from the supplied BOM; order-of-magnitude, assuming a 150 mm, 2 kg cube).

Quantity	Basis / expression	Value	Source
Torque constant K_t	$9.5493/K_v$, $K_v = 380\,\mathrm{rpm/V}$	0.0251 N m/A	Derived (datasheet K_v)
Peak torque	$K_t \times 16\,\mathrm{A}$ (180 s rating)	≈ 0.40 N m	Derived
Continuous torque	$K_t \times 5 - -8\,\mathrm{A}$ (motor-thermal limit)	0.13–0.20 N m	Derived; measure-TBD
Usable wheel speed (6S)	$K_v\,V$; 8436 rpm no-load nominal, derated	≈ 7000 rpm (≈ 733 rad/s)	Derived
Electrical frequency	12 pole pairs × $116.7\,\mathrm{s}^{-1}$ at 7000 rpm	1400 Hz vs 2000 Hz driver ceiling	Derived
Required jump-up momentum	inelastic-collision model, 1.5× margin (edge)	$\approx 0.29\,\mathrm{kg\,m^2/s}$	Derived (estimate)
Target wheel inertia	H/ω at 733 rad/s (edge; corner ≈ 1.7×)	$\geq 3.9 \times 10^{-4}\,\mathrm{kg\,m^2}$	Derived; measure-TBD

Continued on next page

Table 15 (continued)

Quantity	Basis / expression	Value	Source
Stored kinetic energy / wheel	$\frac{1}{2}J\omega^2$ at 733 rad/s	$\approx$ 105 J ($\approx$ 320 J for 3)	Derived

Table 16: Items not covered by the supplied BOM — still to be bought or fabricated. Lead-time risk H/M/L.

Item	Qty	Requirement driving it	Lead-time risk
Reaction-wheel rims	3	Target wheel inertia (Table 15); density-at-radius $\Rightarrow$ laser-/water-cut steel ring, $\approx$ 120 mm OD, 3–4 mm thick	H
Frame stock (Al angle / extrusion / CF tube)	1 set	Three mutually orthogonal, rigid motor faces	H
Spare motor + spare moteus-n1	1 each	Resilience against in-service failure; long resupply lead time	H
E-stop / anti-spark switch + inline fuse	1	Safety: cut energy input on the 40–60 A bus	M
Bench PSU (current-limited, 0–30 V, $\geq$ 5 A)	1	Safe firmware bring-up (never on the battery)	M
LiPo charger + safe bag + cell alarm	1	Battery charging and storage safety	M
Bus bulk capacitor	1	Regen-braking transient ($\approx$ 105 J/wheel); 470–1000 µF, $\geq$ 50 V low-ESR	L
XT30 connector pairs	3+	moteus-n1 power connector (BOM supplies XT90 only)	L
M3 fasteners + heat-set inserts + threadlocker	1 set	Vibration-proof assembly	L
Wire (14 AWG bus, 24–26 AWG signal) + bullets, heat-shrink	1 set	Power and signal harness	L
Filament (PETG, TPU, CF-nylon)	—	All printed structure and brackets	L